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Subject: IDS Roll No.: 21 Exp. No.: 1

Name of the Experiment: _____

Performed on: _____

Submitted on: _____

Marks	Teacher's Signature with date
<u>10</u>	<u>[Signature]</u>

- Q1 Define an algorithm list any 4 characteristics of algorithm
- Soln A Step by Step procedure or formula for solving a problem is known as algorithm characteristics are
- 1 Well defined input: It should procedure accept two or more inputs
 - 2 Well defined output: It should produce at least one output
 - 3 finiteness: The algorithm must terminate after a no. of steps
 - 4 Definiteness: Each step must be clear and unambiguous.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

Q2] Define data structure and Abstract data type (ADT). Give one example of each.

→ A data structure is a way of organizing and storing data so it can be used efficiently.

example :-

Array :- Stores element in contiguous memory location.

Abstract Data Type.

An ADT is a logical model that defines what operation can be performed on data, without specifying how they are implemented.

Example :-

Stack ADT - operation : push, pop, peak.

Q3] Explain the process of converting a problem into program. Illustrate the role of algorithms.

Process.

1. Problem definition - Understand the Problem clearly
2. Analysis - Identify input, output and Constraints.
3. Algorithm design - Create step-by-step Solution.
4. Flowchart / Pseudocode - represent logic visually.
5. Coding - Convert logic into Program.
6. Testing & Debugging - Find and fix errors.
7. Documentation & Maintenance - Record and update.

Role of algorithm.

Algorithm act as a blueprint for the program. They ensure logical correctness before coding. They in writing efficient.

Q4] Explain linear and non-linear data structures with example.

— linear data structures: Data elements are arranged sequentially.
ex: Arrays, linked list, stack, Queue.

— Non linear data structures.

Data elements are not stored sequentially they form hierarchical or network relationships.

ex = Tree, graph.

Q5] Differentiate between algorithm and program, Analyse their roles in SDLC.

Algorithm	Program
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Step by-Step logic	actual Code written in a language
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language independent

language dependent

Focus on Solution

Focus on implementation

Written before coding

Written during coding

Rules in SDLC.

1. Algorithm used in design phase plan Solution

2. Program used in implementation phase to execute.

3. Good algorithm lead to efficient Program

Q6] Relationship among data, data structure and algorithm.

Data, Raw facts or values.

Data structure organizes data

Algorithm processes data

effect on efficiency
Speed

Memory use.

examples : searching in an array
(linear search)

searching in a binary search
Tree $\rightarrow$ faster.